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Enhancing Fluid Characterization and Pore Fluid Content Accuracy Through Integrated Formation Water Resistivity Analysis: A Case Study

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Abstract

Fluid characterization and saturation modeling are foundational to effective field development, particularly in carbonate reservoirs where water chemistry is highly variable and dynamic. This paper presents a case study from a carbonate field in Oman, where uncertainties in determining Formation Water Resistivity (R_w) significantly impacted water saturation estimates and fluid property modeling. To address these challenges, an integrated calibration workflow was developed, combining surface water sample analysis, salinity profiling, and wireline log interpretation. Laboratory measurements—including total dissolved solids, ionic concentrations, and resistivity—were used to assess formation conditions and identify geochemical trends. Special attention was given to salinity variations influenced by hydrodynamic gradients and proximity to disposal wells.

The study revealed a broad range of R_w values across the field, with notable discrepancies between laboratory and log-derived estimates. These differences were most pronounced in wells affected by water disposal activities, where contamination and salinity gradients distorted sample readings. A refined R_w model was constructed through zonal adjustments and contamination screening, resulting in improved alignment between calculated and observed saturation endpoints. This enhanced model significantly reduced uncertainty in hydrocarbon volume forecasts, particularly in low-quality zones with marginal net pay.

Recognizing the evolving nature of formation water chemistry, a time-lapse fluid monitoring program was initiated to track changes in ionic composition over time. This ongoing effort aims to further refine the R_w model and ensure its adaptability to dynamic reservoir conditions. The study demonstrates that a robust and evolving R_w framework not only improves saturation accuracy but also strengthens the reliability of fluid property models, offering a scalable solution for complex carbonate reservoirs and long-term development planning.

This paper demonstrates how R_w calibration can be a critical enabler for accurate fluid property modeling. The new R_w approach bridges the gap between real samples and log data, creating a stronger basis for reserves estimation. The method is adaptable to reservoirs with variable water salinity and hydrodynamic

complexity, offering a field-ready framework to enhance confidence in development planning and long-term production forecasting.

Introduction

Accurate fluid characterization is a cornerstone of successful reservoir management, particularly in carbonate reservoirs where heterogeneity and complex water chemistry pose significant challenges. In such environments, the determination of Formation Water Resistivity (R_w) plays a pivotal role in saturation modeling, which in turn influences fluid property integration and reserves estimation.

Traditional methods of estimating R_w often fall short in reservoirs affected by hydrodynamic gradients, disposal well interference, and variable salinity. This paper introduces an integrated approach to R_w calibration that leverages laboratory measurements, wireline log interpretation, and contamination screening to enhance fluid characterization. The case study presented herein demonstrates the application of this methodology in a carbonate field, highlighting its impact on saturation modeling, fluid input accuracy, and overall field development planning.

The significance of this work lies not only in its technical rigor but also in its practical applicability. As operators increasingly seek to optimize recovery from mature assets and complex reservoirs, the need for accurate and adaptable fluid models becomes more pressing. By addressing the limitations of conventional R_w estimation techniques and offering a scalable solution, this study contributes to the broader goal of enhancing reservoir understanding and maximizing hydrocarbon recovery.

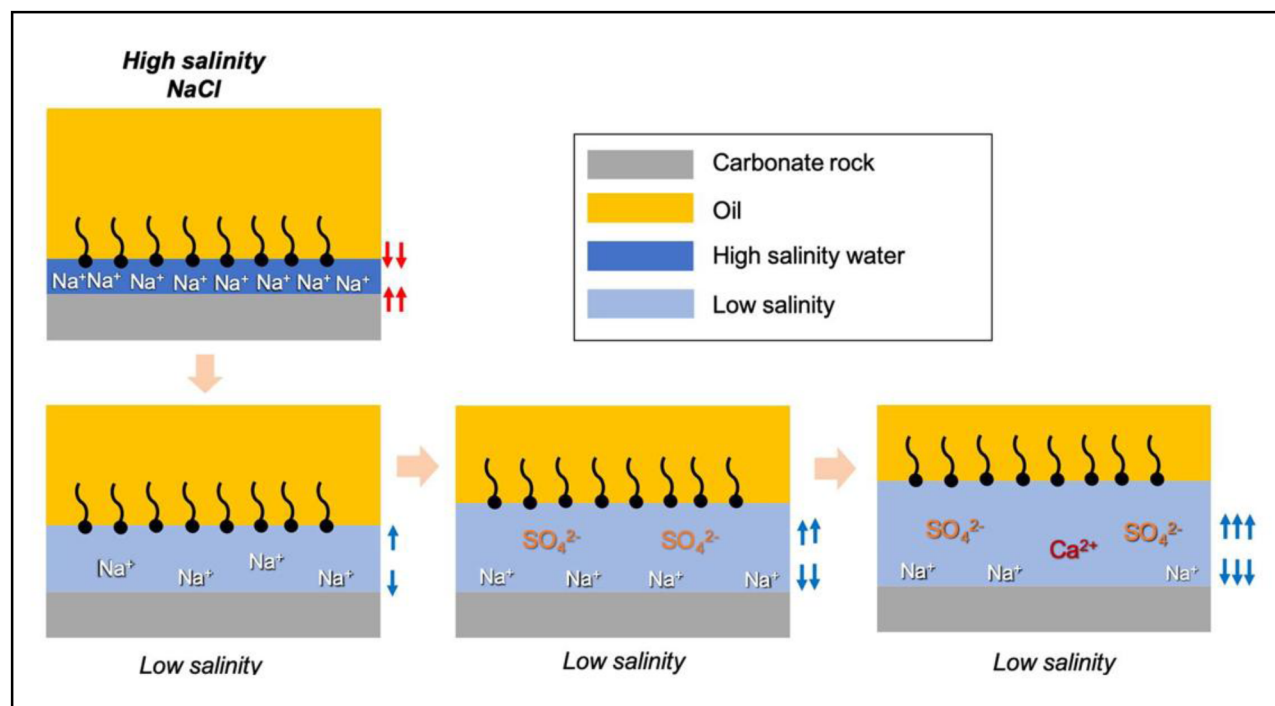


Figure 1—This conceptual diagram highlights how changes in ionic composition—particularly the presence of sodium, calcium, and sulfate ions—impact formation properties such as fluid distribution. These effects are critical to understanding the role of salinity in formation water resistivity and its implications for saturation modeling in carbonate reservoirs (Ref. 1).

Geological and Reservoir Context

The reservoir under study is part of the Carbonate Formation, a widespread shallow marine carbonate platform that developed across Oman and much of the Middle East during the Early Cretaceous. Within the Block A area, deposition occurred in an inner to mid-ramp setting, resulting in a heterogeneous mix of facies.

These include rudist-rich floatstones, *Bacinella*/*Lithocodium*-dominated floatstones and boundstones, and orbitolinoid-rich floatstones—each contributing to the reservoir's complex petrophysical behavior.

The Carbonate Formation is regionally recognized for its prolific oil productivity and intricate depositional architecture. In this study area, the gross pay thickness reaches up to (25-30) meters, with an average porosity of approximately 28-31%. Permeability is highly variable, ranging from less than 10 millidarcies in tight mudstones to over 500 millidarcies in rudist-dominated build-ups. This variability necessitates a detailed understanding of facies distribution and diagenetic overprints to accurately model fluid flow and saturation behavior.

The reservoir is subdivided into six stratigraphic zones, defined through core-based sedimentological analysis and distinctive wireline log responses. This zonation supports consistent geological and dynamic modeling across the field. The primary hydrocarbon-bearing intervals are Zones 1 and 2, which are dominated by rudist-rich carbonates. These zones typically exhibit a combined True Vertical Thickness (TVT) of 6 to 8 meters, though in some locations they can reach up to 10-12 meters, significantly influencing well performance.

A key distinction between Zones 1 and 2 lies in their diagenetic history. Zone 1 is more heavily cemented with calcite and dolomite, resulting in lower porosity values typically between 10% and 20%. In contrast, Zone 2 exhibits higher porosity, ranging from 25% to 33%, making it more favorable for hydrocarbon storage and flow. These differences underscore the importance of integrating geological, petrophysical, and production data when calibrating formation water resistivity and modeling saturation across the reservoir (Ref. 2).

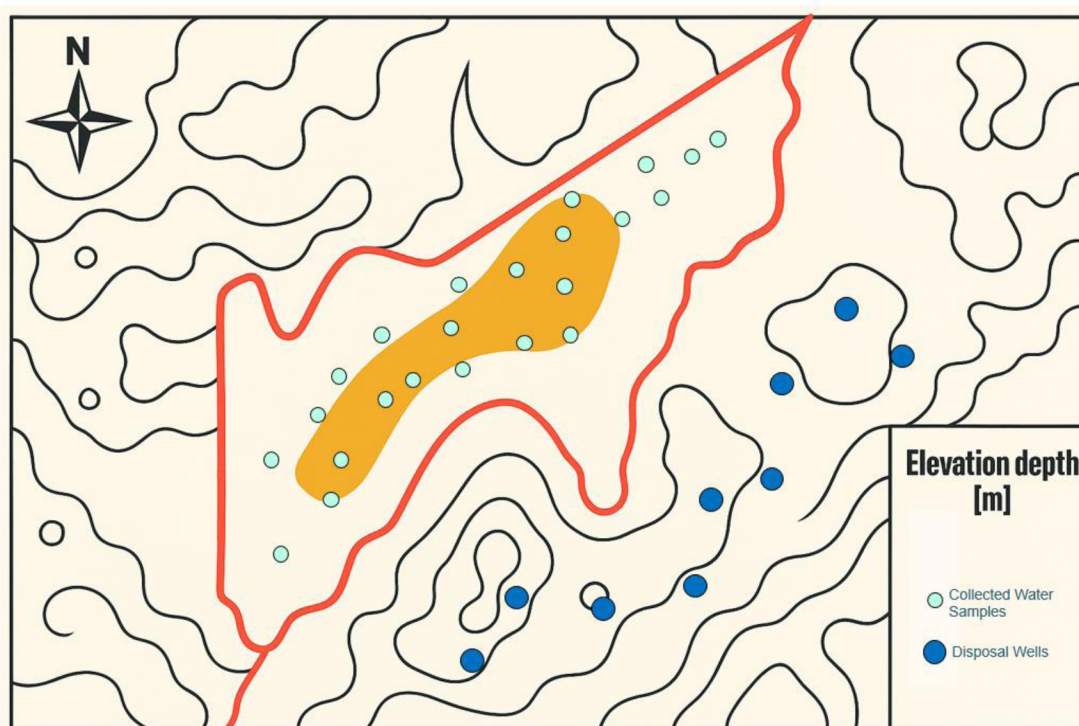


Figure 2—Cartoon-style topographic map highlighting the structural configuration of the carbonate reservoir. Only the central area—outlined in red—is colored to emphasize the primary zone of interest, while surrounding regions are rendered in white. The map showcases a northwest-southeast trending structural high associated with dense well coverage. Light blue circles represent locations where water samples were collected, Dark blue circles denote disposal wells and Black lines indicate producer wells. These elements are clarified in the legend located at the bottom right corner of the map.

Data Acquisition and Analytical Framework

A comprehensive data acquisition and analysis campaign was undertaken to support the calibration of Formation Water Resistivity (R_w) across the carbonate reservoir. The strategy focused on capturing spatial and chemical variability in formation water through surface sampling, followed by rigorous laboratory analysis.

The process begins with sample receiving and sorting, where water samples collected from various wells are cataloged and prepared for analysis. Each sample undergoes decanting to remove residual crude oil, ensuring that the water phase is isolated for accurate chemical testing. Following this, samples are filtered using Whatman filter paper to eliminate particulates and prepare the fluid for further processing.

From this point, the workflow branches into three analytical pathways:

1. **Preservation Pathway:** Samples are acidified to stabilize ionic content and prevent precipitation. These acidified samples are then stored in refrigeration units to maintain integrity until analysis.
2. **Dilution and Instrumental Analysis Pathway:** Samples are diluted, typically at a 10x ratio, to meet the sensitivity requirements of advanced instrumentation. Diluted samples are analyzed using:
 - Electrical Conductivity (EC) measurements to assess ionic strength.
 - Ion Chromatography (IC) for detailed ion profiling.
 - Inductively Coupled Plasma Optical Emission Spectroscopy (ICP-OES) for elemental analysis, particularly for trace metals and salinity-related ions.
3. **Direct Measurement Pathway:** Some samples are subjected to Total Dissolved Solids (TDS) analysis via gravimetry, pH measurement (pH), and titration to quantify specific chemical species such as bicarbonates and chlorides.

This multi-path workflow ensures that each sample is analyzed using the most appropriate method based on its condition and the target parameters. The integration of these techniques provides a robust dataset for salinity profiling, contamination screening, and R_w estimation. The structured approach also allows for cross-validation between methods, increasing confidence in the derived values and supporting the development of a reliable R_w calibration model.

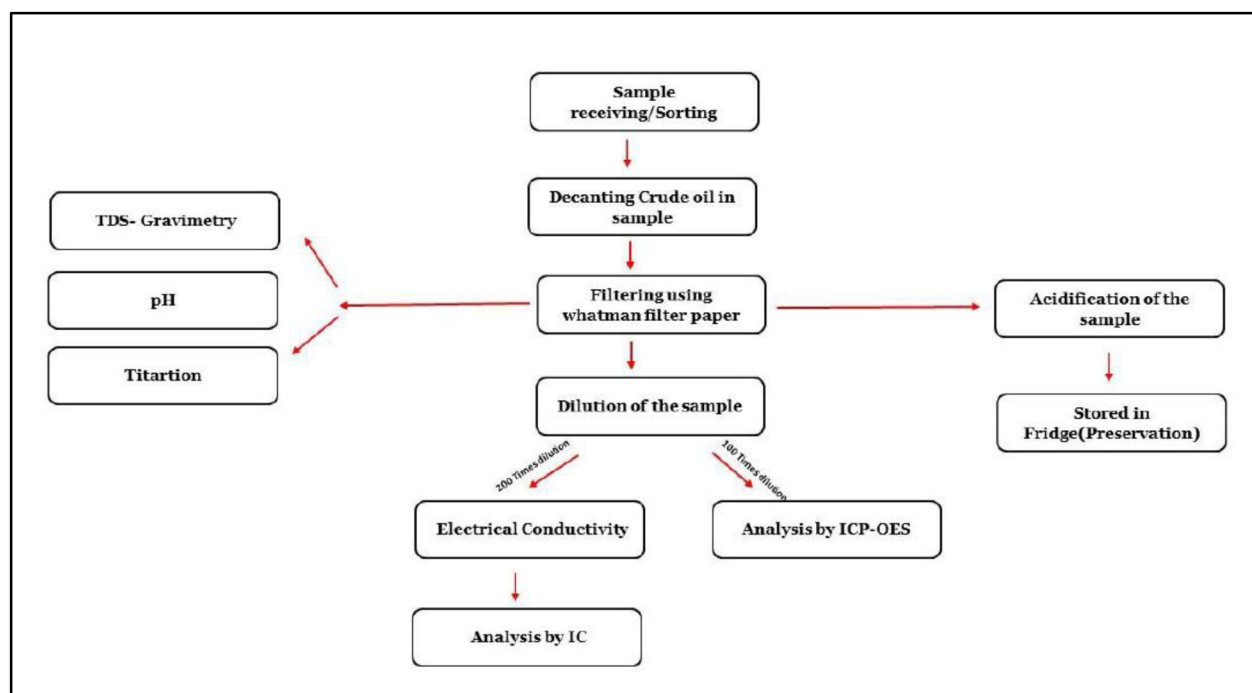


Figure 3—This diagram outlines the complete laboratory workflow from sample reception to final analysis. It includes crude oil decanting, filtration, acidification for preservation, dilution for instrumental analysis (IC, ICP-OES), and direct methods such as TDS gravimetry, pH measurement, and titration. Each pathway contributes to a comprehensive understanding of water chemistry and supports accurate R_w calibration.

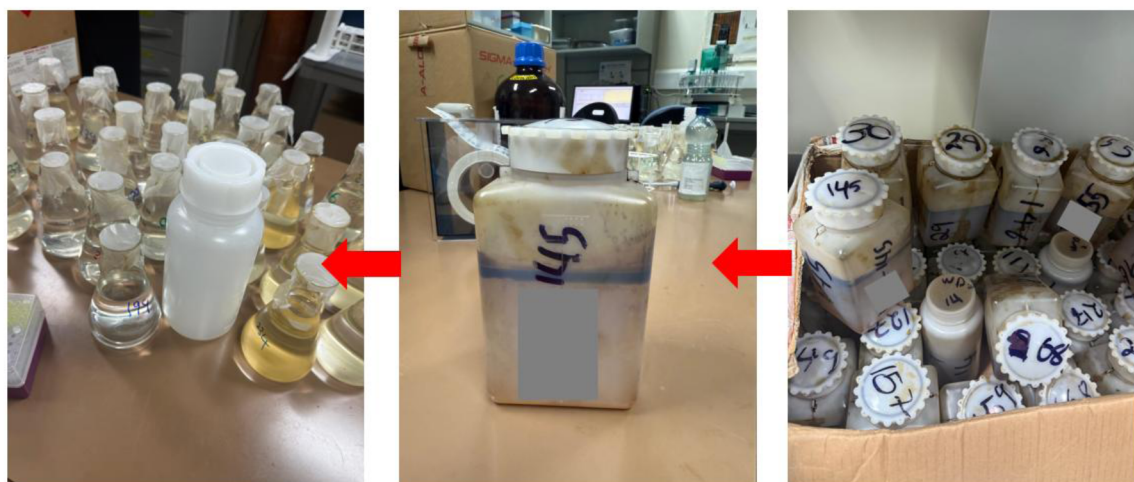


Figure 4—This image shows the progression of sample preparation for laboratory analysis. The right section displays raw samples with visible contamination and labeling, representing the initial condition. The middle section shows an intermediate stage with partial cleaning. The left section features fully cleaned and filtered samples, ready for chemical testing and resistivity analysis.

Formation Water Resistivity (R_w) Calibration Workflow

The R_w calibration process began with a comparative analysis of laboratory-derived resistivity values across multiple wells. Initial discrepancies were observed, particularly in areas near disposal wells or areas with complex facies architecture. To address these inconsistencies, a multi-step calibration strategy was implemented. Salinity profiles were used to identify areas with stable water chemistry, which served as benchmarks for calibration. Samples showing anomalous salinity or chemical signatures were flagged for potential contamination and statistically down-weighted to minimize their impact.

A final R_w model was constructed using a weighted average of laboratory measurements. This model was validated through cross-plot analysis and comparison with historical production data to ensure consistency with observed fluid behavior. The final R_w values were then converted using a standardized method for calculating the resistivity of NaCl water solutions (Figure 5). This approach uses the chemical composition of the solution—specifically the concentration of total dissolved solids—to approximate the sodium chloride content in ppm, allowing for a more representative resistivity estimate.

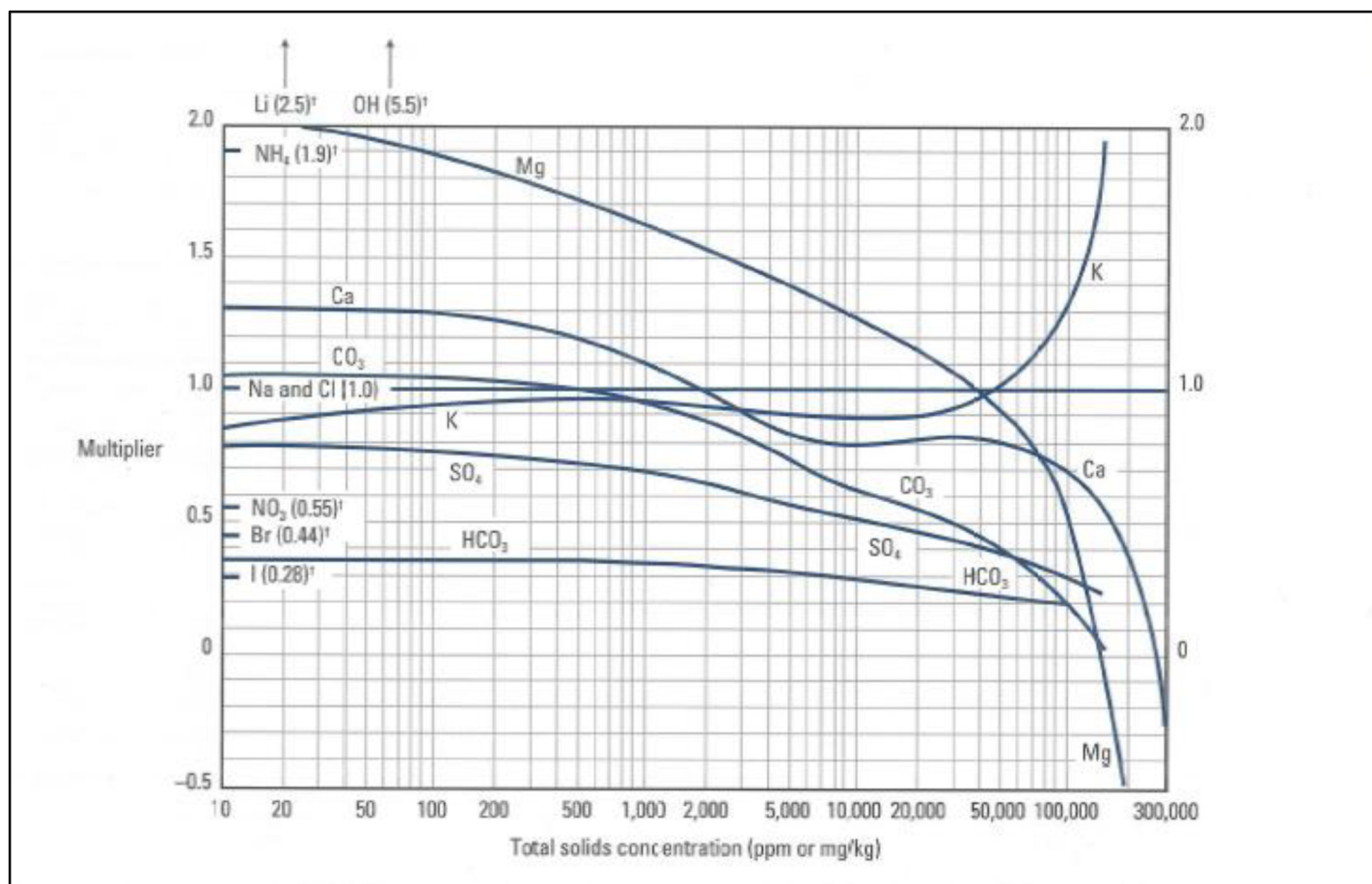


Figure 5—This Schlumberger chart provides a method to convert total solids concentration into an equivalent NaCl concentration in ppm. It uses weighting multipliers for common formation water salts to estimate the resistivity of the solution at a given temperature (Ref.3).

Supporting this workflow was a rigorous laboratory protocol. Water samples were preserved immediately with nitric acid (HNO_3) to maintain integrity. Total dissolved solids (TDS) and total suspended solids (TSS) were measured gravimetrically, while salinity and resistivity were calculated from electrical conductivity. Additional analyses included titration-based determination of bicarbonate and carbonate content, and quantification of sulfate and chloride ions. This integrated approach—combining field data, lab analysis, and geological context—ensures a robust and scalable R_w calibration framework suitable for reservoirs with variable salinity, contamination risks, and complex geology.

Results and Interpretation

To support the calibration of Formation Water Resistivity (R_w), a detailed chemical analysis of formation water samples was conducted, focusing on nine key ionic species: Sodium, Potassium, Calcium, Magnesium, Chloride, Sulphate, Bicarbonate, Strontium, and Nitrate. These ions were selected for their relevance to salinity profiling and their influence on fluid resistivity behavior in carbonate reservoirs.

The temporal distribution of ion concentrations across different sampling batches revealed notable patterns. Sodium and Chloride levels remained relatively stable over time, indicating a consistent salinity profile across the field. In contrast, Bicarbonate and Strontium concentrations showed greater variability, particularly in more recent samples. This dispersion suggests possible geochemical shifts, either due to spatial heterogeneity in the reservoir or evolving conditions influenced by production activities or water-rock interactions.

Statistical analysis of the ionic concentrations provided further insight into the chemical makeup of the formation water. Median values were observed as follows: Sodium at approximately 26,752 ppm, Potassium at 353 ppm, Calcium at 3,347 ppm, Magnesium at 578 ppm, Chloride at 48,142 ppm, Sulphate at 437.4 ppm, Bicarbonate at 134.45 ppm, Strontium at 318.68 ppm, and Nitrate at 7.6 ppm. These central tendencies help characterize the reservoir's baseline water chemistry and identify outliers that may indicate contamination or localized anomalies (Figure 6).



Figure 6—Comparison of Elemental Concentrations in ppm. This figure presents scatter plots and box plots for key ionic species—Sodium, Potassium, Calcium, Magnesium, Chloride, Sulphate, Bicarbonate, Strontium, and Nitrate—measured in parts per million (ppm). Scatter Plots (Left): These illustrate the distribution of elemental concentrations across samples. Color coding indicates the timing of sample collection: Blue: Earlier samples, Orange: Later samples, Green (L): Even later samples, taken after the orange group, while the box plots (right) summarize the statistical spread and central tendencies of each element.

In general, this compositional analysis confirms the saline nature of the formation water, with Sodium and Chloride being the dominant ions. The elevated Bicarbonate and Strontium levels in some samples—particularly in later batches—may reflect interactions with carbonate-rich lithologies or evolving reservoir conditions. These insights are critical for refining the equivalent NaCl conversion and ensuring accurate Rw estimation across zones with varying water chemistries and sampling timelines.

To assess the spatial and statistical behavior of formation water properties, a comprehensive analysis of salinity and water resistivity (Rw) was conducted across the study area. Salinity measurements, expressed in kiloparts per million (kppm), ranged from approximately 71 to 104 kppm. The median salinity was found to be 84 kppm, with the first and third quartiles at 77 and 93 kppm respectively, indicating moderate variability. The average salinity across all sampling batches was approximately 80,467 ppm, and about 80% of the data fell within the range of 72,870 to 88,192 ppm. These values suggest a relatively consistent salinity profile throughout the field, which supports the reliability of zonal calibration benchmarks used in the study.

Water resistivity (R_w) values showed a narrower range, spanning from 0.034 to 0.045 ohm-m. The median R_w was approximately 0.0405 ohm-m, with an average of 0.040 ohm-m. The interquartile range, between 0.038 and 0.042 ohm-m, reflects low variability across the dataset. Notably, around 80% of the R_w values were concentrated between 0.036 and 0.043 ohm-m. A consistent declining trend in R_w was observed toward the northeastern section of the field, which may be indicative of a geochemical gradient or the influence of structural and hydrodynamic factors (Figure 7).

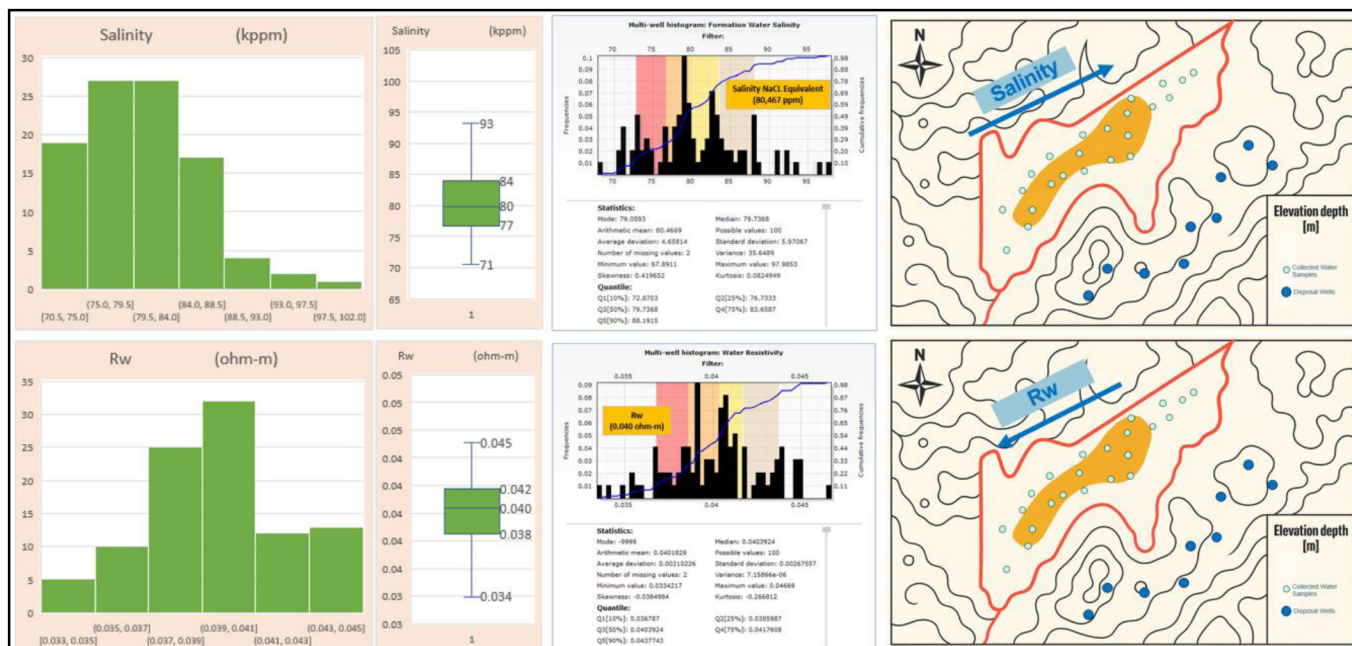


Figure 7—Spatial and Statistical Distribution of Salinity and Water Resistivity (R_w). This figure presents bar charts, box plots, and spatial maps illustrating the distribution and variability of salinity and R_w across the study area. It highlights the inverse relationship between salinity and resistivity and reveals consistent geochemical trends across sampling batches.

The observed inverse relationship between salinity and R_w not only validates the calibration workflow but also reinforces the utility of salinity as a reliable proxy for resistivity estimation in zones lacking direct measurements. This correlation, supported by both statistical and spatial coherence, enhances confidence in the zonal modeling of water properties. More importantly, it sets the stage for understanding how these calibrated inputs influence broader reservoir evaluation workflows. The implications of this refined R_w model extend directly into fluid property modeling, where even small variations in input parameters can significantly affect saturation estimates and hydrocarbon volume calculations.

Impact on Fluid Property Modeling

Accurate modeling of fluid properties in the reservoir requires careful calibration of input parameters and recognition of areal variability. The analysis of hydrocarbon saturation across multiple reservoir units revealed that formation water resistivity (R_w) plays a critical role in saturation estimation, contributing a significant portion of the total error. Despite its relatively low absolute value, R_w showed a high sensitivity in the model, with a notable negative error percentage. Porosity and true resistivity (R_t) also contributed to the overall uncertainty, though to a lesser extent. These findings underscore the importance of precise R_w calibration and error quantification when modeling hydrocarbon volumes.

Petrophysical uncertainty analyses using partial derivatives revealed that water saturation (S_w) uncertainty can reach up to 8.2%, depending on the sensitivity of input parameters. Conversely, saturation uncertainty derived from range-based calculations of formation water resistivity (R_w) showed a strong

dependency on porosity. In high-porosity zones, the uncertainty may be as low as 6%, whereas in lower-quality zones, it can increase to 10% (Figure 8).

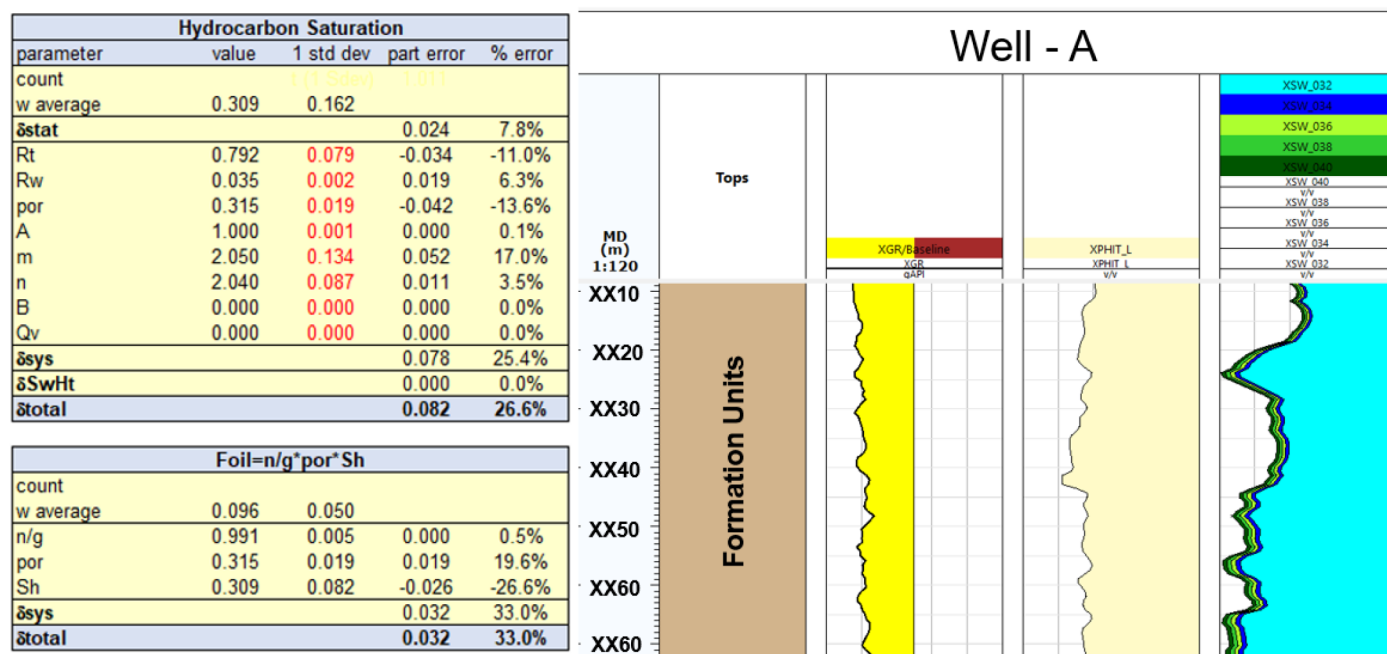


Figure 8—Spatial and Statistical Distribution of Salinity and Water Resistivity (Rw). This figure presents bar charts, box plots, and spatial maps illustrating the distribution and variability of salinity and Rw across the study area. It highlights the inverse relationship between salinity and resistivity and reveals consistent geochemical trends across sampling batches.

Discussion

The results of this study underscore the complexity of fluid characterization in carbonate reservoirs, particularly when dealing with variable salinity and dynamic formation water chemistry. While the calibration of Formation Water Resistivity (Rw) has clearly improved saturation modeling, the process revealed several layers of uncertainty that merit deeper consideration.

One of the most striking observations is the spatial variability in salinity and R_w across the field. The consistent increase in salinity toward the northeastern area, coupled with a corresponding decrease in R_w , suggests a geochemical gradient likely influenced by structural highs, hydrodynamic flow, or legacy injection activities. However, the lack of a clear trend in injector salinity complicates this interpretation. It raises questions about the mixing behavior of injected fluids and the extent to which they alter native water chemistry.

From a modeling standpoint, the sensitivity of R_w to saturation calculations is both a strength and a vulnerability. On one hand, it allows for fine-tuned adjustments that improve hydrocarbon volume estimates. On the other, it introduces a disproportionate source of error in low-porosity zones, where even minor inaccuracies in R_w can lead to significant miscalculations in water saturation. This highlights the need for zone-specific calibration and error propagation analysis, particularly in marginal pay areas where development decisions hinge on subtle petrophysical differences.

The temporal variability in ionic concentrations—especially Potassium and Calcium—adds another layer of complexity. These shifts may reflect evolving reservoir conditions, such as mineral dissolution, water-rock interaction, or production-induced changes. The initiation of a time-lapse fluid monitoring program is a commendable step, but its success will depend on consistent sampling protocols and integration with dynamic reservoir models.

In fact, while the integrated R_w calibration approach has delivered measurable improvements, it also exposes the reservoir's sensitivity to both spatial and temporal fluid variations. Future work should focus on expanding the dataset in geochemically complex zones, refining zonal modeling techniques, and leveraging machine learning to detect subtle patterns in water chemistry. Only through such iterative and multidisciplinary efforts can we hope to build truly resilient fluid models for complex carbonate systems.

Conclusions and Recommendations

This case study illustrates the critical role of Formation Water Resistivity (R_w) calibration in enhancing fluid characterization and saturation modeling in carbonate reservoirs. The integrated workflow—combining laboratory measurements, log interpretation, and contamination screening—provides a robust framework for R_w estimation.

Spatial analysis revealed a general increase in salinity toward the northeastern area of the field, with an average salinity of approximately 80,467 ppm. Around 80% of the values fall between 72,870 and 88,192 ppm. In contrast, R_w values tend to decrease in the same direction, with ~80% ranging between 0.036 and 0.043 ohm-m. While injector salinity shows no consistent trend, these spatial variations underscore the need for targeted data acquisition.

To strengthen the reliability of the R_w model and improve fluid characterization, it is recommended to collect additional data from the northeastern area, with particular attention to trend variations in the coming Batches (Fluid Analysis Campaigns). Continued monitoring of water composition over time is essential, especially focusing on changes in Potassium and Calcium concentrations, which may signal evolving reservoir conditions. Furthermore, maintaining multidisciplinary collaboration and iterative refinement of the R_w model will ensure its adaptability and robustness across dynamic reservoir environments.

What makes this study distinctive is its comprehensive sampling strategy, which spans the entire field and incorporates multiple timeframes. This approach enables the ability to monitor temporal changes in water composition, offering a more dynamic and responsive understanding of reservoir behavior.

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